

Pradip Sapkota

(507) 469-8488 | pradipsapkota.ce@outlook.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Minnesota State University, Mankato

B.S. Computer Engineering, GPA: 3.7/4.0

Mankato, MN

Expected Dec. 2026

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB, SystemVerilog, VHDL, SQL, Ladder Logic

Hardware & Embedded: FPGA, Microcontrollers, Raspberry Pi, Sensors, Digital Logic, PLC, Modbus TCP

Engineering Tools: KiCad, Autodesk EAGLE, Oscilloscope, Spectrum Analyzer, AutoCAD, SolidWorks, ModelSim

Testing & Systems: Hardware Debugging, System Validation, Linux, Git/GitHub, AWS Lambda, DynamoDB

EXPERIENCE

Ruppell Inc.

Software Engineering Associate

Jul. 2025 – Aug. 2026

Mankato, MN

- Developed industrial analytics dashboards using React, AWS Lambda, and DynamoDB to visualize sensor readings and system performance.
- Improved data-processing and API performance by **40%** through Python workflows and AWS API Gateway optimization.
- Collaborated with engineering teams on requirements, system integration, functional testing, defect resolution, and deployment readiness.

Minnesota State University, Mankato

IT Solutions Consultant

Jan. 2024 – Present

Mankato, MN

- Diagnose hardware, software, network, authentication, and system-integration failures across major desktop and mobile platforms.
- Perform root-cause analysis to isolate device, connectivity, configuration, permission, and enterprise-application failures.
- Support deployment, configuration, validation, and documentation while resolving **50+ technical incidents**.

Minnesota State University, Mankato

Undergraduate Research Assistant – Edge-AI Digital Twin

Aug. 2023 – Jan. 2024

Mankato, MN

- Supported digital-twin research integrating real-time sensor data, embedded computing, and cyber-physical system monitoring.
- Assisted with Python development, data processing, testing, and technical research for intelligent control experiments.
- Researched edge computing and predictive maintenance and contributed code, technical findings, and documentation.

PROJECTS

Smart Door Access Control System | KiCad, Autodesk EAGLE, Embedded Systems, Python

Apr. 2025

- Designed and fabricated a PCB for an embedded access-control system integrating electronic locks, sensors, and authentication hardware.
- Developed hardware–software control logic with anti-spoofing, fail-safe behavior, and reliability testing.

Interactive AI Voice Assistant | KiCad, Autodesk EAGLE, Raspberry Pi, Python

Dec. 2024

- Designed and fabricated a PCB supporting a Raspberry Pi voice-assistant system with audio, GPIO-controlled hardware, and external AI services.
- Performed hardware–software integration and iterative debugging to improve system reliability and response behavior.

LEADERSHIP

Institute of Electrical and Electronics Engineers (IEEE)

Events Lead

Aug. 2023 – Present

Mankato, MN

- Lead embedded-design competitions and technical workshops, coordinating schedules, equipment, logistics, and faculty/student teams.